

OUARZAZATE, Morocco

WMO: 602650

Lat: 30.939N

Lon: 6.909W

Elev: 3783

StdP: 12.79

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	32.3	35.3	1.3	6.7	58.1	5.3	8.3	60.5	33.8	56.9	29.6	54.0	2.6	300	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.5	102.6	61.3	101.8	61.2	100.1	60.9	66.2	88.5	64.7	88.3	63.6	88.8	11.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.4	90.3	71.1	57.4	81.0	71.5	55.0	74.1	72.7	33.0	88.5	31.8	87.7	30.8	87.9	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.9	23.2	19.3	DB	28.0	105.4	3.1	1.5	25.7	106.5	23.9	107.4	22.2	108.3	19.9	109.4	(4)
			WB	22.9	68.9	3.2	2.7	20.6	70.9	18.8	72.5	17.0	74.1	14.7	76.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	68.1	49.7	53.8	60.3	67.0	74.0	81.7	87.5	85.6	77.8	69.1	58.4	50.8
	DBStd	13.71	4.53	4.65	4.98	5.51	5.78	4.13	2.75	3.56	4.55	4.65	5.52	3.89
	HDD50	124	62	18	1	0	0	0	0	0	0	0	5	38
	HDD65	1661	476	314	157	43	7	0	0	0	18	205	441	
	CDD50	6714	51	125	322	511	744	952	1161	1102	835	591	258	62
	CDD65	2774	0	0	12	104	285	502	696	637	386	144	8	0
	CDH74	38452	2	15	277	1248	3543	7003	10986	9202	4588	1452	136	0
	CDH80	21693	0	0	38	349	1643	4092	7163	5662	2332	404	10	0
Wind	WSAvg	6.3	4.8	5.7	7.1	8.3	8.3	7.3	7.2	6.8	6.3	5.3	4.9	3.9
Precipitation	PrecAvg	3.5	0.4	0.4	0.5	0.2	0.2	0.1	0.1	0.4	0.5	0.6	0.6	0.3
	PrecMax	8.4	3.0	2.2	3.9	1.1	1.8	0.4	0.6	1.3	2.6	3.3	3.6	1.8
	PrecMin	0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.7	0.6	0.6	0.7	0.3	0.4	0.1	0.1	0.4	0.6	0.8	0.9	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.0	76.5	84.1	89.7	96.7	102.3	105.6	104.1	99.0	90.0	81.3	71.6
		MCWB	48.6	49.4	52.3	55.4	57.9	60.0	62.1	61.3	61.5	57.4	54.7	49.2
	2%	DB	69.3	73.1	80.3	86.4	94.0	100.2	103.9	102.2	96.3	87.4	77.8	68.1
		MCWB	47.4	48.7	51.5	54.7	57.4	59.6	61.6	61.7	59.8	57.0	53.3	47.8
	5%	DB	66.0	70.0	77.0	84.2	91.6	98.3	102.3	100.5	94.1	84.4	74.9	65.6
		MCWB	46.4	47.6	50.5	53.9	57.3	59.5	61.2	61.3	59.2	56.4	52.3	47.4
	10%	DB	62.7	67.2	73.7	81.0	89.2	95.8	100.7	98.7	91.5	82.1	71.6	62.5
		MCWB	44.7	46.7	49.6	52.7	56.7	58.9	61.0	61.3	59.0	55.9	51.1	46.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.5	53.7	56.2	58.7	64.3	64.6	67.1	68.9	66.2	62.4	58.0	54.1
		MCDB	67.6	66.0	71.2	81.6	84.9	88.6	95.3	89.2	82.5	77.3	74.5	63.3
	2%	WB	49.2	51.4	53.9	56.7	61.2	63.0	65.3	66.6	64.2	60.5	55.9	51.3
		MCDB	64.4	64.6	72.0	80.2	84.8	87.8	94.2	87.8	82.6	75.2	71.2	61.4
	5%	WB	47.4	49.8	52.3	55.2	59.4	61.9	63.8	65.3	62.8	59.2	54.0	49.5
		MCDB	62.7	64.6	71.5	79.2	83.4	87.6	94.9	87.3	83.1	74.7	69.8	60.5
	10%	WB	45.8	48.1	51.0	54.0	58.0	60.9	62.6	64.0	61.5	57.9	52.4	47.8
		MCDB	60.7	63.9	70.1	77.7	82.9	87.5	95.1	87.2	82.5	75.3	68.5	60.0
Mean Daily Temperature Range	5% DB	MDBR	25.9	25.3	24.8	24.7	25.5	26.8	27.5	26.0	25.4	23.8	24.7	24.7
		MCDBR	30.9	30.4	29.7	27.9	28.4	29.1	29.2	28.1	28.7	27.2	28.3	28.9
		MCWBR	16.1	14.5	12.4	10.3	10.0	9.9	9.8	9.0	9.6	10.3	12.9	14.7
	5% WB	MCDBR	27.7	25.4	25.7	25.6	25.3	25.7	26.2	24.2	23.9	22.4	24.4	23.8
		MCWBR	14.8	12.2	11.1	10.3	10.0	8.8	9.4	8.6	8.4	9.2	11.6	12.6
Clear-Sky Solar Irradiance	taub	0.263	0.305	0.368	0.388	0.429	0.475	0.562	0.531	0.435	0.387	0.316	0.282	
	taud	2.601	2.456	2.234	2.199	2.080	1.957	1.751	1.843	2.102	2.245	2.480	2.560	
	Ebn at Noon	313	305	289	287	274	260	238	244	265	271	287	299	
	Edh at Noon	25	32	43	46	52	59	72	65	48	39	28	25	
All-Sky Solar Radiation	RadAvg	1243	1518	1894	2222	2400	2529	2386	2154	1904	1541	1265	1122	
	RadStd	73	109	101	104	136	139	106	88	121	82	86	70	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.90	N/A	-4.00	+1.12	-1.37	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)		+0.86	N/A	-3.33	+1.21	-1.08	N/A	N/A	N/A	+275	+308

Nomenclature: See separate page